C2.3 How do metals and non-metals combine to form compounds?		
Teaching and learning narrative	**Assessable learning outcomes** *Learners will be required to:*	***Linked learning opportunities***
Group 0 contains elements with a full outer shell of electrons. This arrangement is linked to their inert, unreactive properties. They exist as single atoms and hence are gases with low melting and boiling points. Group 1 elements combine with Group 7 elements by ionic bonding. This involves a transfer of electrons leading to charged ions. Atoms and ions can be represented using dot and cross diagrams as simple models (IaS3). Metals, such as Group 1 metals, lose electrons from the outer shell of their atoms to form ions with complete outer shells and with a positive charge. Non-metals, such as Group 7, form ions with a negative charge by gaining electrons to fill their outer shell. The number of electrons lost or gained determines the charge on the ion. The properties of ionic compounds such as Group 1 halides can be explained in terms of the ionic bonding. Positive ions and negative ions are strongly attracted together and form giant lattices. Ionic compounds have high melting points in comparison to many other substances due to the strong attraction between ions which means a large amount of energy is needed to break the attraction between the ions. They dissolve in water because their charges allow them to interact with water molecules. They conduct electricity when molten or in solution because the charged ions can move, but not when solid because the ions are held in fixed positions.	1. recall the simple properties of Group 0 including their low melting and boiling points, their state at room temperature and pressure and their lack of chemical reactivity 2. explain how observed simple properties of Groups 1, 7 and 0 depend on the outer shell of electrons of the atoms and predict properties from given trends down the groups 3. explain how the reactions of elements are related to the arrangement of electrons in their atoms and hence to their atomic number 4. explain how the atomic structure of metals and non-metals relates to their position in the Periodic Table 5. describe the nature and arrangement of chemical bonds in ionic compounds 6. explain ionic bonding in terms of electrostatic forces and transfer of electrons 7. calculate numbers of protons, neutrons and electrons in atoms and ions, given atomic number and mass number or by using the Periodic Table M1a 8. construct dot and cross diagrams for simple ionic substances	**Practical work:** • test the properties of ionic compounds **Ideas about Science:** • dot and cross diagrams as models of atoms and ions, and the limitations of these models (IaS3) • 2-D and 3-D representations as simple models of the arrangement of ions, and the limitations of these models (IaS3)

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The arrangement of ions can be represented in both 2-D and 3-D. These representations are simple models which have limitations, for example they do not fully show the nature or movement of the electrons or ions, the interaction between the ions, their arrangement in space, their relative sizes or scale (IaS3).	9.	explain how the bulk properties of ionic materials are related to the type of bonds they contain	
	10.	use ideas about energy transfers and the relative strength of attraction between ions to explain the melting points of ionic compounds compared to substances with other types of bonding	
	11.	describe the limitations of particular representations and models of ions and ionically bonded compounds, including dot and cross diagrams, and 3-D representations	
	12.	translate information between diagrammatic and numerical forms and represent three dimensional shapes in two dimensions and vice versa when looking at chemical structures for ionic compounds M4a, M5b	